Ground-Truth Question Set

Lookup Category

1. When was the Taxes folder created?
2. What is the most recent event in the timeline, and for which location?
3. When was the USB drive's most recently recorded activity?
4. What is the path of the most recently created folder, and when was it created?

Summary Category

5. What activity occurred on USB drives?
6. What happened between [start time] and [end time] (inclusive)?
7. Was any network share accessed, and if so, what was accessed on it?
8. Summarize the user's activity in chronological order.

Unusual Activity Category

9. Is there any activity that would be unusual for a normal user and worth an investigator's attention?
10. Were any system folders accessed?
11. Are there any events close together in time (within about a minute) that might be related?
12. Were any folders with sensitive-sounding names created or accessed?

Counting Category

13. How many folders on the USB drive appear in the timeline (including the drive's root folder)?
14. How many folders have events between [start time] and [end time] (inclusive)?
15. How many folders under Documents appear in the timeline (including nested folders)?
16. How many drives or network locations other than C: appear?

Unanswerable Category

17. What files were inside the Taxes folder?
18. Was any data copied to or from the USB drive?
19. How many times was the Taxes folder opened?
20. When was the USB drive disconnected, and how long was it connected?

NOTE: The following answers for all three timelines assume that the timelines are being viewed in (UTC-05:00) Eastern Daylight Time.

Control Timeline Sample Answers

Lookup Category

1. **When was the Taxes folder created?**

   No folder named Taxes was recorded in the timeline.

2. **What is the most recent event in the timeline, and for which location?**

   The most recent event in the timeline was an Item Last Registry Write event in Desktop at 8:06 PM on 10/27/2026.

3. **When was the USB drive's most recently recorded activity?**

   No USB drive activity was recorded in the timeline.

4. **What is the path of the most recently created folder, and when was it created?**

   The most recently created folder was Documents\Recipes\Dinner at 7:57 PM on 10/27/2026.

Summary Category

5. **What activity occurred on USB drives?**

   No USB drive activity was recorded in the timeline.

6. **What happened between 7:55 and 8:00pm (inclusive)?**

   Summary of recorded activity 7:55 – 8:00PM (UTC-5:00) Eastern Daylight Time on 10/27/2026

   1. *7:55 PM: Documents\Recipes was created and accessed*
   2. *7:57 PM: Documents\Recipes\Dinner was created and accessed.*
   3. *8:00 PM: Pictures was accessed.*

   This is normal browsing and folder creation in the user's own profile.

7. **Was any network share accessed, and if so, what was accessed on it?**

   No network share was recorded in the timeline.

8. **Summarize the user's activity in chronological order.**

   Summary of recorded activity on 10/27/2026 in (UTC-5:00) Eastern Daylight Time

   1. *7:52 PM: Documents\School was created and accessed.*
   2. *7:53 PM: Documents\School\Notes was created and accessed.*
   3. *7:55 PM: Documents\Recipes was created and accessed.*
   4. *7:57 PM: Documents\Recipes\Dinner was created and accessed.*
   5. *8:00 PM: Pictures was accessed.*
   6. *8:02 PM: Music was accessed.*
   7. *8:05 PM: Downloads was accessed.*
   8. *8:06 PM: Desktop was accessed.*

Overall, the activity recorded is normal file browsing and folder creation in the user's own profile.

Unusual Activity Category

9. **Is there any activity that would be unusual for a normal user and worth an investigator's attention?**
   None of the recorded activity would be unusual for a normal user or worth an investigator's attention.
10. **Were any system folders accessed?**
    No system folders were recorded in the timeline.
11. **Are there any events close together in time (within about a minute) that might be related?**
    No recorded events were close enough together in time to suggest a relation.
12. **Were any folders with sensitive-sounding names created or accessed?**
    No folders with sensitive-sounding names were recorded in the timeline.

Counting Category

13. **How many folders on the USB drive appear in the timeline (including the drive's root folder)?**
    No USB drive activity was recorded in the timeline.
14. **How many folders have events between 7:52 and 7:57 PM (inclusive)?**
    Five folders have events between 7:52 and 7:57 PM on 10/27/2026.
    1. *Documents\School\Notes*
    2. *Documents\School*
    3. *Documents\Recipes*
    4. *Documents*
    5. *Documents\Recipes\Dinner*
15. **How many folders under Documents appear in the timeline (including nested folders)?**
    Four folders under Documents appear in the timeline.
    1. *School*
    2. *School\Notes*
    3. *Recipes*
    4. *Recipes\Dinner*
16. **How many drives or network locations other than C: appear?**
    No drives or network locations other than C: were recorded in the timeline.

Unanswerable Category

17. **What files were inside the Taxes folder?**

    The files inside of a folder cannot be determined from shellbags data.

18. **Was any data copied to or from the USB drive?**

    Whether or not data was copied to or from an external drive cannot be determined from shellbags data.

19. **How many times was the Taxes folder opened?**

    The number of folder accesses cannot be determined from shellbags data.

20. **When was the USB drive disconnected, and how long was it connected?**

    Shellbags data cannot determine when external drives are disconnected nor how long they are connected.

Small Timeline Sample Answers

Lookup Category

1. **When was the Taxes folder created?**

   Documents\Taxes was created at 7:04 PM on 10/27/2026.

2. **What is the most recent event in the timeline, and for which location?**

   The most recent event in the timeline was an Item Last Registry Write event in Downloads at 7:34 PM on 10/27/2026.

3. **When was the USB drive's most recently recorded activity?**

   The most recent activity recorded on the USB drive was an Item Last Registry Write event in E:\Receipts at 7:24 PM on 10/27/2026.

4. **What is the path of the most recently created folder, and when was it created?**

   The most recently created folder was E:\Receipts at 7:24 PM on 10/27/2026.

Summary Category

5. **What activity occurred on USB drives?**

   Summary of recorded activity on E: on 10/27/2026 in (UTC-5:00) Eastern Daylight Time

   1. *7:19 PM: E:\Backup was created and accessed.*
   2. *7:21 PM: E:\Photos was created and accessed.*
   3. *7:24 PM: E:\Receipts was created and accessed.*

   Three folders were created and accessed on E:.

6. **What happened between 7:18 and 7:24 PM (inclusive)?**

   Summary of recorded activity 7:18 – 7:24 PM (UTC-5:00) Eastern Daylight Time on 10/27/2026

   1. *7:19 PM: Documents was accessed. E:\Backup was created and accessed.*
   2. *7:21 PM: E:\Photos was created and accessed.*
   3. *7:24 PM: E:\Receipts was created and accessed.*

   The activity centers on the USB drive. E:\Backup was accessed about 14 seconds after Documents was revisited.

7. **Was any network share accessed, and if so, what was accessed on it?**

   No network share activity was recorded in the timeline.

8. **Summarize the user's activity in chronological order.**

   Summary of recorded activity on 10/27/2026 in (UTC-5:00) Eastern Daylight Time

   1. *2:03 PM: C:\Windows was modified.*
   2. *6:28 PM: C:\Windows\System32 was accessed and modified.*
   3. *7:04 PM: Documents\Taxes was created and accessed.*

4. *7:06 PM: Documents\School and Documents\School\Notes were created and accessed.*
5. *7:08 PM: Pictures was accessed.*
6. *7:19 PM: Documents was accessed. E:\Backup was created and accessed.*
7. *7:21 PM: E:\Photos was created and accessed.*
8. *7:24 PM: E:\Receipts was created and accessed.*
9. *7:25 PM: C:\Windows was accessed.*
10. *7:29 PM: C:, C:\Windows, and C:\Windows\System32 were accessed.*
11. *7:34 PM: Downloads was accessed.*

Overall, the user created several folders in their own profile and on a USB drive. Windows system folders were later accessed.

Unusual Activity Category

9. **Is there any activity that would be unusual for a normal user and worth an investigator's attention?**

   Yes, Windows system folders were modified and accessed on 10/27/2026.

   1. *2:03 PM: C:\Windows was modified.*
   2. *6:28 PM: C:\Windows\System32 was accessed and modified.*
   3. *7:25 PM: C:\Windows was accessed.*
   4. *7:29 PM: C:, C:\Windows, and C:\Windows\System32 were accessed.*

10. **Were any system folders accessed?**

    Yes, Windows system folders were modified and accessed on 10/27/2026.

    1. *2:03 PM: C:\Windows was modified.*
    2. *6:28 PM: C:\Windows\System32 was accessed and modified.*
    3. *7:25 PM: C:\Windows was accessed.*
    4. *7:29 PM: C:, C:\Windows, and C:\Windows\System32 were accessed.*

11. **Are there any events close together in time (within about a minute) that might be related?**

    Yes, there are two pairs of events that happened within a minute of each other on 10/27/2026.

    1. *7:19 PM: Documents was accessed. E:\Backup was created and accessed.*
    2. *7:24 – 7:25 PM: E:\Receipts was created and accessed. C:\Windows was accessed.*

12. **Were any folders with sensitive-sounding names created or accessed?**

    Yes. C:\Windows and C:\Windows\System32 were accessed and modified. Documents\Taxes was created and accessed.

Counting Category

13. **How many folders on the USB drive appear in the timeline (including the drive's root folder)?**

    Four folders on the USB drive appear in the timeline.

    1. *E:\Backup*
    2. *E:\Photos*
    3. *E:\Receipts*
    4. *E:*

14. **How many folders have events between 7:19 and 7:24 PM (inclusive)?**

    Five folders have events between 7:19 and 7:24 PM on 10/27/2026.

    1. *Documents*
    2. *E:\Backup*
    3. *E:\Photos*
    4. *E:\Receipts*
    5. *E:*

15. **How many folders under Documents appear in the timeline (including nested folders)?**

    Three folders under Documents appear in the timeline.

    1. *Taxes*
    2. *School*
    3. *School\Notes*

16. **How many drives or network locations other than C: appear?**

    One USB drive (E:) appears in the timeline. No network shares appear.

Unanswerable Category

17. **What files were inside the Taxes folder?**

    The files inside of a folder cannot be determined from shellbags data.

18. **Was any data copied to or from the USB drive?**

    Whether or not data was copied to or from an external drive cannot be determined from shellbags data.

19. **How many times was the Taxes folder opened?**

    The number of folder accesses cannot be determined from shellbags data.

20. **When was the USB drive disconnected, and how long was it connected?**

    Shellbags data cannot determine when external drives are disconnected nor how long they are connected.

Large Timeline Sample Answers

Lookup Category

1. **When was the Taxes folder created?**
   Documents\Taxes was created at 4:11 PM on 10/28/2026
2. **What is the most recent event in the timeline, and for which location?**
   The most recent event in the timeline was an Item Last Registry Write event in Desktop at 5:17 PM on 10/28/2026.
3. **When was the USB drive's most recently recorded activity?**
   The most recent activity recorded on the USB drive was an Item Last Registry Write event in E:\Receipts at 4:50 PM on 10/28/2026.
4. **What is the path of the most recently created folder, and when was it created?**
   The most recently created folder was E:\Receipts at 4:50 PM on 10/28/2026.

Summary Category

5. **What activity occurred on USB drives?**
   E:\Photos was created and accessed at 7:21 PM on 10/27/2026.
   Summary of recorded activity on E: on 10/28/2026 in (UTC-5:00) Eastern Daylight Time
   1. *4:43 PM: E:\Photos\Vacation was created and accessed.*
   2. *4:46 PM: E:\Backup was created and accessed.*
   3. *4:48 PM: E:\Photos\2026 was created and accessed.*
   4. *4:50 PM: E:\Receipts was created and accessed.*

   Four folders were created and accessed on E:.
6. **What happened between 4:24 and 5:00 PM (inclusive)?**
   Summary of recorded activity 4:24 – 5:00 PM (UTC-5:00) Eastern Daylight Time on 10/28/2026
   1. *4:24 PM: Network and \\VBOXSVR\tl_large were accessed.*
   2. *4:43 PM: E:\Photos\Vacation was created and accessed.*
   3. *4:46 PM: E:\Backup was created and accessed.*
   4. *4:48 PM: E:\Backup\2026 was created and accessed.*
   5. *4:50 PM: E:\Receipts was created and accessed.*

   A network share (\\VBOXSVR\tl_large) was accessed, then four folders were created and accessed on E:.
7. **Was any network share accessed, and if so, what was accessed on it?**
   Yes, \\VBOXSVR was accessed and tl_large was accessed on it.

8. **Summarize the user's activity in chronological order.**

Summary of recorded activity on 10/27/2026 in (UTC-5:00) Eastern Daylight Time

1. *2:03 PM: C:\Windows was modified.*
2. *7:21 PM: E:\Photos was created and accessed.*

Summary of recorded activity on 10/28/2026 in (UTC-5:00) Eastern Daylight Time

1. *3:12 PM: C:\Windows\System32 was modified.*
2. *3:40 PM: Documents\Projects was created and accessed.*
3. *3:42 PM: Documents\Projects\Project01 was created and accessed.*
4. *3:44 PM: Documents\Projects\Project02 was created and accessed.*
5. *3:47 PM: Documents\Projects\Project02\Drafts was created and accessed.*
6. *3:49 PM: Documents\Projects\Project03 was created and accessed.*
7. *3:51 PM: Music was accessed.*
8. *3:53 PM: Documents\School was created and accessed.*
9. *3:56 PM: Documents\School\Notes was created and accessed.*
10. *3:58 PM: Documents\School\Notes\2026 was created and accessed.*
11. *4:00 PM: Documents\School\Notes\2026\Birthday was created.*
12. *4:01 PM: Documents\School\Notes\2026\Birthday was accessed.*
13. *4:03 PM: Documents\Projects\Project04 was created and accessed.*
14. *4:09 PM: Documents\Passwords was created and accessed. C:\Windows was accessed.*
15. *4:11 PM: Documents\Taxes was created and accessed.*
16. *4:13 PM: Documents\Backup was created and accessed.*
17. *4:15 PM: C:\Windows\System32 was accessed.*
18. *4:24 PM: Network and \\VBOXSVR\tl_large were accessed.*
19. *4:43 PM: E:\Photos\Vacation was created and accessed.*
20. *4:46 PM: E:\Backup was created and accessed.*
21. *4:48 PM: E:\Backup\2026 was created and accessed.*
22. *4:50 PM: E:\Receipts was created and accessed.*
23. *5:01 PM: Documents was accessed.*
24. *5:06 PM: C:, C:\Windows, and C:\Windows\System32 were accessed.*
25. *5:17 PM: Desktop was accessed.*

On 10/27/2026, C:\Windows was modified and E:\Photos was created. The rest of the activity occurred on 10/28/2026. System folders were modified, then four folders were created under Documents\Projects. Music was accessed, then three nested folders were created under Documents\School. Another folder was then created under Documents\Projects, and three more

folders were created under Documents. A system folder was accessed, then two network locations were accessed. Four folders were then created on E:, and then Documents was accessed. Two system folders were accessed again, and then Desktop was accessed.

Unusual Activity Category

9. **Is there any activity that would be unusual for a normal user and worth an investigator's attention?**

   Yes, C:\Windows was modified at 2:03 PM on 10/27/2026.

   Windows system folders were modified and accessed on 10/28/2026.

   1. *3:12 PM: C:\Windows\System32 was modified.*
   2. *4:09 PM: C:\Windows was accessed.*
   3. *4:15 PM: C:\Windows\System32 was accessed.*
   4. *5:06 PM: C:, C:\Windows, and C:\Windows\System32 were accessed.*

10. **Were any system folders accessed?**

    Yes, C:\Windows was modified at 2:03 PM on 10/27/2026.

    Windows system folders were modified and accessed on 10/28/2026.

    1. *3:12 PM: C:\Windows\System32 was modified.*
    2. *4:09 PM: C:\Windows was accessed.*
    3. *4:15 PM: C:\Windows\System32 was accessed.*
    4. *5:06 PM: C:, C:\Windows, and C:\Windows\System32 were accessed.*

11. **Are there any events close together in time (within about a minute) that might be related?**

    Yes. Documents\Passwords was created then C:\Windows was accessed within a second. Documents\Passwords was accessed 2 seconds later.

12. **Were any folders with sensitive-sounding names created or accessed?**

    Yes. C:\Windows and C:\Windows\System32 were modified and accessed. Documents\Passwords and Documents\Taxes were created and accessed.

Counting Category

13. **How many folders on the USB drive appear in the timeline (including the drive's root folder)?**

    Six folders on the USB drive appear in the timeline.

    1. *E:\Photos*
    2. *E:\Photos\Vacation*
    3. *E:\Backup*

4. *E:\Backup\2026*
5. *E:\Receipts*
6. *E:*

14. **How many folders have events between 3:42 and 3:49 PM (inclusive)?**

    Four folders have events between 3:42 and 3:49 PM on 10/28/2026.

    1. *Documents\Projects\Project01*
    2. *Documents\Projects\Project02*
    3. *Documents\Projects\Project02\Drafts*
    4. *Documents\Projects\Project03*

15. **How many folders under Documents appear in the timeline (including nested folders)?**

    13 folders under Documents appear in the timeline.

    1. *Projects*
    2. *Projects\Project01*
    3. *Projects\Project02*
    4. *Projects\Project02\Drafts*
    5. *Projects\Project03*
    6. *School*
    7. *School\Notes*
    8. *School\Notes\2026*
    9. *School\Notes\2026\Birthday*
    10. *Projects\Project04*
    11. *Passwords*
    12. *Taxes*
    13. *Backup*

16. **How many drives or network locations other than C: appear?**

    One USB drive (E:) and one network share (\\VBOXSVR) appear in the timeline.

Unanswerable Category

17. **What files were inside the Taxes folder?**

    The files inside of a folder cannot be determined from shellbags data.

18. **Was any data copied to or from the USB drive?**

    Whether or not data was copied to or from an external drive cannot be determined from shellbags data.

19. **How many times was the Taxes folder opened?**

The number of folder accesses cannot be determined from shellbags data.

20. **When was the USB drive disconnected, and how long was it connected?**

    Shellbags data cannot determine when external drives are disconnected nor how long they are connected.